Is physics in crisis?

“Physics is in crisis” – the phrase turns up in books, on YouTube and in the science pages of newspapers. Since the Higgs boson was found in 2012, the big accelerators have found no new physics. Dark matter has still not been seen in any laboratory, the expansion rate of the universe does not agree with itself, and after forty years string theory has no tested predictions. At the same time physics has never been more precise or more useful. This page sets out what the critics say, what the counter-arguments are, and where things stand in the autumn of 2026 – with two interactive figures.

What is a crisis in science?

The word has a specific meaning in the philosophy of science. In *The Structure of Scientific Revolutions* (1962) Thomas Kuhn described how scientists spend most of their time working within a paradigm, solving “puzzles”. A crisis arises when anomalies pile up and can no longer be explained away – and it ends in a revolution in which a new paradigm takes over (see What is knowledge?).

Physics has been in crisis before, and the crises were fruitful:

Around 1900

In 1900 Lord Kelvin spoke of “two clouds” over classical physics: the ether could not be detected, and heat radiation and heat capacities did not fit the theory. One cloud became relativity, the other quantum mechanics.

The 1920s

The “old quantum theory” was a patchwork. The crisis was resolved by Heisenberg, Schrödinger and Dirac in 1925–28 (see Quantum mechanics).

Around 1930

The electrons from beta decay did not have a fixed energy. Niels Bohr considered abandoning conservation of energy; Pauli instead proposed an invisible particle – the neutrino, detected in 1956.

The 1960s

The particle zoo and the infinities of quantum field theory led many to give up on field theory. The answer turned out to be quarks and gauge theories – the Standard Model (see Particle physics).

The present situation is unusual because it is not mainly a matter of theories failing. On the contrary: the Standard Model and the standard model of cosmology fit almost every measurement. The problem is that we know they cannot be the whole truth – and that nature has not given us the clues we expected.

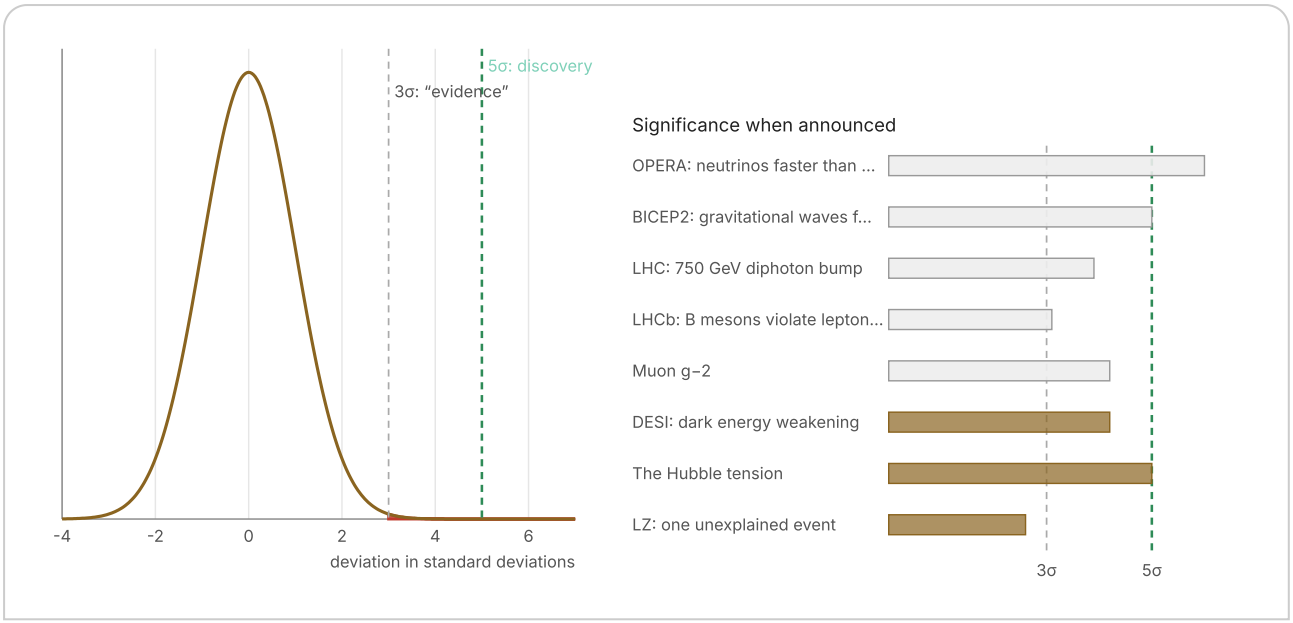
Particle physics after the Higgs

The Higgs boson was found at CERN on 4 July 2012 with exactly the properties the Standard Model predicted. It was a triumph – but also the beginning of a problem. Many theorists had expected the LHC also to find *supersymmetric* particles or other new phenomena, because otherwise the Higgs mass requires a suspiciously fine tuning of the constants of nature (the *hierarchy problem* or *naturalness* argument). After more than ten years of data taking at energies up to 13.6 TeV, nothing has been found. This has been called the “nightmare scenario”: the Higgs and nothing else.

The coming High-Luminosity LHC will collect about ten times as much data, and the LHC will run until 2041. On 22 May 2026 the CERN Council updated the European Strategy for Particle Physics and named the electron–positron collider FCC-ee, in a 91 km tunnel, as the preferred next flagship project; the final decision is to be taken by 2028 and depends on funding and public consultations in France and Switzerland. Critics, among them Sabine Hossenfelder, argue that many billions should be spent differently when there is no clear prediction to test. Supporters reply that a “Higgs factory” will measure the properties of the Higgs particle with a precision that can reveal new physics indirectly, whatever it may be.

Anomalies that went away

In particle physics a discovery requires 5 sigma: the probability that a random fluctuation produces so large an effect must be less than 1 in 3.5 million. That sounds strict, but history shows why it is needed. Over the past fifteen years a series of promising deviations from the Standard Model have appeared and then vanished again.



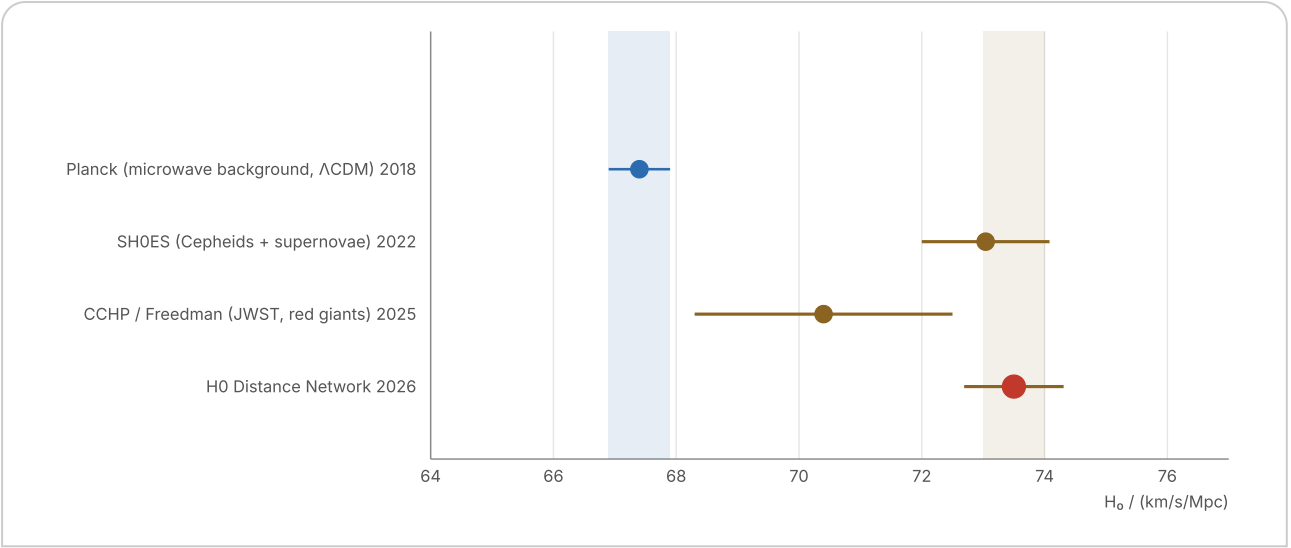
The most talked-about was the magnetic moment of the muon, $g - 2$. In 2021 Fermilab’s measurement deviated by 4.2 sigma from the theoretical prediction of the time. In June 2025 the final result came with a precision of 127 parts per billion – but by then lattice-QCD calculations had changed the theoretical prediction so that it now agrees with the measurement. The tension now lies between two ways of calculating the theory, not between theory and experiment.

Tensions in cosmology

The standard model of cosmology, ΛCDM, describes the cosmic microwave background, the distribution of galaxies and the formation of the elements with only six parameters. But it says that only about 5 % of the universe is ordinary matter, 27 % dark matter and 68 % dark energy – two things we do not understand.

Dark matter has been detected through its gravity, but not in any laboratory. The xenon detector LZ beneath South Dakota has set the world’s tightest limits on WIMP particles. In September 2026 LZ reported a single event that is hard to explain with known backgrounds – but with a significance of only 2.6 sigma, and the scientists stress that they are not claiming to have seen dark matter.

The Hubble tension. The expansion rate of the universe H_0 can be calculated from the cosmic microwave background with ΛCDM, or measured directly with a distance ladder of Cepheids and supernovae. The two methods give different answers, and the difference has become clearer the more precise the measurements have become.



Changing dark energy? In 2024 and 2025 the galaxy survey DESI found signs that dark energy is not Einstein's constant Λ but weakens with time – at a significance of between 2.8 and 4.2 sigma depending on which other data it is combined with. DESI completed its five-year survey in 2026 with more than 47 million galaxies and quasars, and the first results from the full data set are expected in 2027. If the hint holds up, it will be the first break with ΛCDM.

Quantum gravity and string theory

Quantum mechanics and general relativity each work superbly, but they cannot be united. Since 1984 string theory has been the leading candidate for a unified theory, but it has produced no prediction that can be tested with existing technology, and it has a "landscape" of some 10^{500} possible universes. Lee Smolin (*The Trouble with Physics*, 2006) and Peter Woit (*Not Even Wrong*, 2006) criticised the way the theory has dominated posts and funding. Neil Turok, then director of the Perimeter Institute, told incoming students in a welcome speech that the situation was a deep crisis and string theory a catastrophe. In 2014 George Ellis and Joe Silk warned in *Nature* against abandoning the requirement that theories must be testable. See String theory – and its critics.

The foundations of quantum mechanics

A hundred years after Heisenberg's breakthrough, physicists still disagree about what quantum mechanics says about reality. When *Nature* surveyed more than 1100 researchers in 2025, 36 % preferred the Copenhagen interpretation, 15 % the many-worlds interpretation and 7 % Bohm's pilot-wave theory – but only 24 % thought their preferred interpretation was actually true. Some see this as a crisis of understanding; others as a philosophical question that does not affect physics. See What does quantum mechanics mean? and Schrödinger's cat.

A sociological crisis?

In *Lost in Math* (2018) Sabine Hossenfelder argued that theorists have let themselves be guided by aesthetic criteria – beauty, naturalness, symmetry – that nature has not confirmed. She herself prefers the word *stagnation* to crisis: the data are well explained by the existing models, but the predictions of fundamental theory have been wrong for decades, and the incentives of the field reward popular topics over risky new ideas. She points out that a failed prediction is often simply moved on to the next generation of experiments.

Others point to working conditions: temporary contracts, pressure to publish many papers, experiments with thousands of authors in which it is hard to be noticed, and a media culture in which every 3-sigma deviation is presented as a possible revolution. Criticism also comes from the opposite direction: many physicists think the crisis rhetoric – especially on social media – damages trust in science and gives a distorted picture of a field that is doing well.

The counter-arguments

Physics is more than fundamental physics

Only a small fraction of physicists work on particles and cosmology. Condensed matter, quantum technology, astrophysics, biophysics and climate physics are enjoying a golden age (see Solid state physics and Laser physics).

Great discoveries

The Higgs boson (2012), gravitational waves (2015), the image of a black hole (2019), neutrino masses and thousands of exoplanets – this is not a stagnant science.

Null results are knowledge

That there is no supersymmetry at LHC energies is a result that has ruled out many theories. The Michelson–Morley experiment was also a null result.

Breakthroughs take time

From Maxwell's equations to radio took 30 years, from Dirac's equation to the positron four, from the Higgs prediction to its discovery 48. Forty years without a breakthrough in quantum gravity is not unique.

What next?

There is no agreement on the way forward, but there are many paths. Precision measurements – atomic clocks, the electron's dipole moment, antihydrogen – can reveal new physics at far higher energies than any accelerator. Gravitational waves, neutrino telescopes and the great sky surveys (DESI, Euclid, the Vera Rubin Observatory) use the universe as a laboratory. Small tabletop experiments test gravity between quantum systems. And the next generation of accelerators can measure the Higgs particle, which is still the most enigmatic part of the Standard Model.

Perhaps the most accurate thing to say is that the foundations of physics are in a period in which nature is not giving us the answers we hoped for – and in which we must therefore ask whether we have been asking the right questions. Whether one calls it a crisis, stagnation or a test of patience depends on temperament. Kelvin's two clouds in 1900 were not a sign that physics was finished either, but that it was about to begin again.

Further reading

- Sabine Hossenfelder: *Lost in Math. How Beauty Leads Physics Astray* (Basic Books 2018)
- Lee Smolin: *The Trouble with Physics* (2006)
- Peter Woit: *Not Even Wrong* (2006)
- George Ellis and Joe Silk: "Scientific method: Defend the integrity of physics", *Nature* 516 (2014)
- CERN: "The CERN Council decided to update the European Strategy for Particle Physics" (22 May 2026)
- Physics World: "Muon g−2 achieves record precision but theoretical tensions remain" (2025)
- Thomas S. Kuhn: *The Structure of Scientific Revolutions* (1962)

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